

Exercises 3 - Machinery Principles

1. A ferromagnetic core is shown in Figure 1. The depth of the core is 5 cm. The other dimensions of the core are as shown in the figure. Find the value of the current that will produce a flux of 0.003 Wb. With this current, assume that the relative permeability of the core is 1000.
 - a. What is the flux density at the top of the core?
 - b. What is the flux density at the right side of the core?

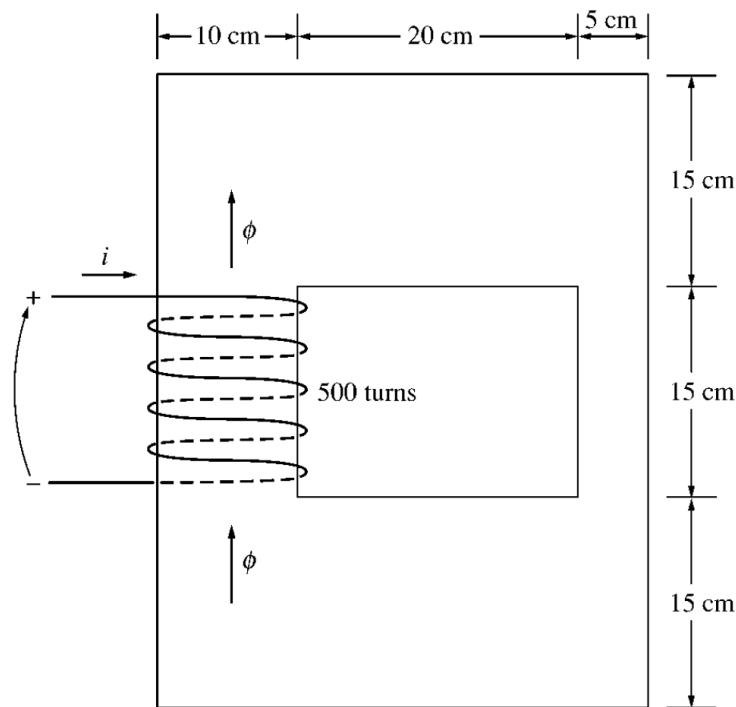


Figure 1

2. A ferromagnetic core with a relative permeability of 2000 is shown in Figure 2. The dimensions are as shown in the diagram, and the depth of the core is 7 cm. The air gaps on the left and right sides of the core are 0.05 cm and 0.07 cm, respectively. If there are 300 turns in the coil wrapped around the center leg of the core and if the current in the coil is 1.0 A, what is the flux of the center the core? What is the flux density?

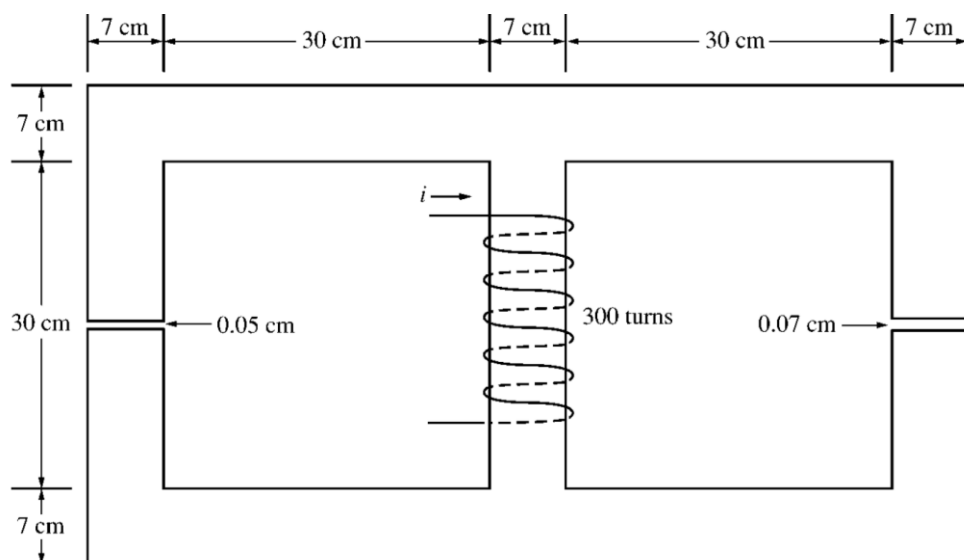


Figure 2